SAFETY DATA SHEET

GULLIVER® (ГУЛЛИВЕР®)

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	17.10.2025	50000058	Date of first issue: 17.10.2025

1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product name	:	GULLIVER® (ГУЛЛИВЕР®)

Manufacturer or supplier's details

Company	:	FMC IP Technology LLC
Address	:	Vernier Branch Chemin de Blandonnet 8 1214 Vernier Switzerland
Telephone	:	+41 22 518 89 61
Emergency telephone number	:	+44 20 3885 0382 (CHEMTREC's European Regional Toll-Free Number) 1 703 / 741-5970 (CHEMTREC - International) 1 703 / 527-3887 (CHEMTREC - Alternate)
Medical Emergency Number	:	+1 651 / 632-6793 (outside USA & Canada, collect call)
E-mail address	:	SDS-Info@fmc.com

Recommended use of the chemical and restrictions on use

Recommended use	:	Can be used as herbicide only.
Restrictions on use	:	Use as recommended by the label.

2. HAZARDS IDENTIFICATION

GHS Classification

Acute toxicity (Dermal)	:	Category 5
Skin irritation	:	Category 3
Short-term (acute) aquatic hazard	:	Category 1
Long-term (chronic) aquatic hazard	:	Category 1

GHS-Labelling

Hazard pictograms :

Signal word	:	Warning
Hazard statements	:	H313 May be harmful in contact with skin.

H316 Causes mild skin irritation.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary statements :

Prevention:

P273 Avoid release to the environment.

Response:

P312 Call a POISON CENTER/ doctor if you feel unwell.
P332 + P313 If skin irritation occurs: Get medical advice/ attention.
P391 Collect spillage.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

Components

Chemical name	CAS-No.	Classification	MAC value mg/m3 / TSEL value	Concentration (% w/w)
azimsulfuron (ISO)	120162-55-2	Acute Tox.5; H313 Aquatic Acute1; H400 Aquatic Chronic1; H410	No data available	>= 30 - < 50
Talc (Mg3H2(SiO3)4)	14807-96-6	No data available	No data available	>= 30 - < 50
Lignosulfonic acid, Sodium salt	8061-51-6	No data available	MPC-STEL: 2 mg/m3 Class 3 - Moderately dangerous Data Source: RU OEL	>= 1 - < 10
magnesium carbonate	546-93-0	Acute Tox.5; H303 Aquatic Acute3; H402	MPC-STEL: 10 mg/m3 Class 4 - Low hazard Data Source: RU OEL	>= 0,25 - < 1
sodium benzoate	532-32-1	Acute Tox.5; H303	MPC-STEL: 5 mg/m3	>= 0,1 - < 0,25

		Acute Tox.5; H313 Eye Irrit.2A; H319 Aquatic Acute3; H402	Class 3 - Moderately dangerous Data Source: RU OEL	

For explanation of abbreviations see section 16.

4. FIRST AID MEASURES

General advice	:	Move out of dangerous area. Show this safety data sheet to the doctor in attendance. Do not leave the victim unattended.
If inhaled	:	Remove to fresh air. If unconscious, place in recovery position and seek medical advice. If experiencing any discomfort, immediately remove from exposure. Light cases: Keep person under surveillance. Get medical attention immediately if symptoms develop. Serious cases: Get medical attention immediately or call for an ambulance.
In case of skin contact	:	If on clothes, remove clothes. If on skin, rinse well with water. Wash off with soap and plenty of water. Get medical attention if irritation develops and persists.
In case of eye contact	:	Flush eyes with water as a precaution. Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist.
If swallowed	:	Keep respiratory tract clear. Do not give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. If symptoms persist, call a physician. Do not induce vomiting without medical advice.
Most important symptoms and effects, both acute and delayed	:	May be harmful in contact with skin. Causes mild skin irritation.
Protection of first-aiders	:	Avoid inhalation, ingestion and contact with skin and eyes.
Notes to physician	:	Treat symptomatically. Immediate medical attention is required in case of ingestion.

5. FIREFIGHTING MEASURES

Suitable extinguishing media	:	Dry chemical, CO2, water spray or regular foam. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media	:	Do not spread spilled material with high-pressure water streams. High volume water jet
Specific hazards during fire-fighting	:	Do not allow run-off from fire fighting to enter drains or water courses.
Hazardous combustion products	:	Fire may produce irritating, corrosive and/or toxic gases. Nitrogen oxides (NOx) Sulphur oxides Carbon oxides Hydrogen cyanide
Specific extinguishing methods	:	Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
Special protective equipment for firefighters	:	Wear self-contained breathing apparatus for firefighting if necessary.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Use personal protective equipment. Avoid dust formation. Avoid breathing dust. If it can be safely done, stop the leak. Remove all sources of ignition. Ensure adequate ventilation. Never return spills in original containers for re-use. Mark the contaminated area with signs and prevent access to unauthorized personnel. Only qualified personnel equipped with suitable protective equipment may intervene. For disposal considerations see section 13.
Environmental precautions	:	Prevent product from entering drains. Prevent further leakage or spillage if safe to do so. If the product contaminates rivers and lakes or drains inform respective authorities.
Methods and materials for containment and cleaning up	:	Never return spills in original containers for re-use. Pick up and transfer the spilled material to a properly labeled container without creating dust. For spills on concrete or other non-porous surfaces, the area can be cleaned using a small quantity of soap and water. Do not allow the cleaning solution to enter drains. Use an inert absorbent material to soak up the cleaning solution and transfer it to the properly labeled container. When the spill occurs on soil, the only effective way to decontaminate the area is to remove the top 5 to 7 centimeters of soil. Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

Advice on protection against fire and explosion	:	Avoid dust formation. Provide appropriate exhaust ventilation at places where dust is formed.
Advice on safe handling	:	Avoid contact with skin and eyes. For personal protection see section 8. Smoking, eating and drinking should be prohibited in the application area. Dispose of rinse water in accordance with local and national regulations.
Conditions for safe storage	:	Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Electrical installations / working materials must comply with the technological safety standards.
Further information on storage conditions	:	Store in closed, labelled containers. The storage room should be constructed of incombustible material, closed, dry, ventilated and with impermeable floor, without access of unauthorised persons or children. The room should only be used for storage of chemicals. Food, drink, feed and seed should not be present. A hand wash station should be available.
Further information on storage stability	:	No decomposition if stored and applied as directed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Personal protective equipment

Respiratory protection	:	In the case of dust or aerosol formation use respirator with an approved filter.
Filter type	:	Particulates type
Hand protection		
Material	:	Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.
Remarks	:	The suitability for a specific workplace should be discussed with the producers of the protective gloves.
Eye protection	:	Eye wash bottle with pure water Tightly fitting safety goggles
Skin and body protection	:	Dust impervious protective suit Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Protective measures	:	Plan first aid action before beginning work with this product. Always have on hand a first-aid kit, together with proper instructions. Wear suitable protective equipment. Ensure that eye flushing systems and safety showers are located close to the working place. When using do not eat, drink or smoke. In the context of professional plant protection use as recommended, the end user must refer to the label and the instructions for use.
Hygiene measures	:	Avoid contact with skin, eyes and clothing. Do not breathe dust. When using do not eat or drink. When using do not smoke. Wash hands before breaks and at the end of workday.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	solid
Form	:	granules
Colour	:	off-white
Odour	:	Faint odour
Odour Threshold	:	not determined
pH	:	5,7 Concentration: 1 %
Melting point/freezing point	:	No data available
Boiling point/boiling range	:	No data available
Flash point	:	Not applicable
Evaporation rate	:	Not applicable
Self-ignition	:	not auto-flammable
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower	:	No data available

flammability limit		
Vapour pressure	:	No data available
Relative vapour density	:	Not applicable
Relative density	:	0,5973
Density	:	No data available
Bulk density	:	No data available
Solubility(ies)		
Water solubility	:	dispersible
Solubility in other solvents	:	No data available
Partition coefficient: n-octanol/water	:	No data available
Auto-ignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity		
Viscosity, dynamic	:	No data available
Viscosity, kinematic	:	Not applicable
Explosive properties	:	Not explosive
Oxidizing properties	:	The product is not oxidizing.
Surface tension	:	42,6 mN/m, 20 °C
Metal corrosion rate	:	Not corrosive to metals
Particle size	:	No data available

10. STABILITY AND REACTIVITY

Reactivity	:	No decomposition if stored and applied as directed.
Chemical stability	:	No decomposition if stored and applied as directed.
Possibility of hazardous reactions	:	No decomposition if stored and applied as directed. Dust may form explosive mixture in air.
Conditions to avoid	:	Avoid extreme temperatures Avoid dust formation.

Heat, flames and sparks.

Incompatible materials : Avoid strong acids, bases, and oxidizers

Hazardous decomposition products : No hazardous decomposition products are known.

11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation
Skin contact

Acute toxicity

May be harmful in contact with skin.

Product:

Acute oral toxicity : LD50 (Rat, male and female): > 5.000 mg/kg
Method: OECD Test Guideline 401
Assessment: The substance or mixture has no acute oral toxicity

Acute inhalation toxicity : Remarks: No data available

Acute dermal toxicity : LD50 (Rat): > 2.000 mg/kg
Method: OECD Test Guideline 402
Assessment: The component/mixture is minimally toxic after single contact with skin.
Remarks: no mortality

Components:

azimsulfuron (ISO):

Acute oral toxicity : LD50 (Rat, male and female): > 5.000 mg/kg
Method: OECD Test Guideline 401
GLP: yes
Assessment: The substance or mixture has no acute oral toxicity

Acute inhalation toxicity : LC50 (Rat): > 5,94 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
GLP: yes
Assessment: The substance or mixture has no acute inhalation toxicity
Remarks: no mortality

Acute dermal toxicity : LD50 (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
GLP: yes
Assessment: The component/mixture is minimally toxic after single contact with skin.

Remarks: no mortality

Talc (Mg3H2(SiO3)4):

Acute oral toxicity	:	LD0 (Rat, male): > 5.000 mg/kg Method: OECD Test Guideline 423 Remarks: no mortality
Acute inhalation toxicity	:	LC0 (Rat, male and female): > 2,1 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Remarks: no mortality
Acute dermal toxicity	:	LD0 (Rat, male and female): > 2.000 mg/kg Method: OECD Test Guideline 402 Remarks: no mortality

Lignosulfonic acid, Sodium salt:

Acute oral toxicity	:	LD50 (Mouse): 6.030 mg/kg

magnesium carbonate:

Acute oral toxicity	:	LD50 (Rat, female): > 2.000 mg/kg Method: OECD Test Guideline 420

sodium benzoate:

Acute oral toxicity	:	LD50 (Rat, male and female): 3.450 mg/kg
Acute inhalation toxicity	:	LC0 (Rat, male and female): > 12,2 mg/l Exposure time: 4 h Test atmosphere: dust/mist Remarks: no mortality Based on data from similar materials
Acute dermal toxicity	:	LD50 (Rabbit, male and female): > 2.000 mg/kg

Skin corrosion/irritation

Causes mild skin irritation.

Product:

Species	:	Rabbit
Assessment	:	Not classified as irritant
Method	:	OECD Test Guideline 404
Result	:	Mild skin irritation

Components:

azimsulfuron (ISO):

Species	:	Rabbit
Method	:	OECD Test Guideline 404
Result	:	No skin irritation
GLP	:	yes

Remarks	:	Information source: Internal study report

Talc (Mg3H2(SiO3)4):

Species	:	reconstructed human epidermis (RhE)
Result	:	No skin irritation

Lignosulfonic acid, Sodium salt:

Remarks	:	May cause skin irritation and/or dermatitis.

magnesium carbonate:

Species	:	reconstructed human epidermis (RhE)
Method	:	OECD Test Guideline 431
Result	:	No skin irritation

sodium benzoate:

Species	:	Rabbit
Method	:	OECD Test Guideline 404
Result	:	No skin irritation

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Species	:	Rabbit
Result	:	No eye irritation
Assessment	:	Not classified as irritant
Method	:	OECD Test Guideline 405
Remarks	:	Minimal effects that do not meet the threshold for classification.
Remarks	:	Product dust may be irritating to eyes, skin and respiratory system.

Components:

azimsulfuron (ISO):

Species	:	Rabbit
Result	:	No eye irritation
Method	:	OECD Test Guideline 405
GLP	:	yes
Remarks	:	Information source: Internal study report

Talc (Mg3H2(SiO3)4):

Species	:	Rabbit
Result	:	No eye irritation
Method	:	OECD Test Guideline 405

Lignosulfonic acid, Sodium salt:

Remarks	:	May irritate eyes.

magnesium carbonate:

Species	:	Rabbit
Result	:	No eye irritation
Method	:	OECD Test Guideline 405

sodium benzoate:

Species	:	Rabbit
Result	:	Eye irritation
Method	:	OECD Test Guideline 405

Respiratory or skin sensitisation

Skin sensitisation

Based on available data, the classification criteria are not met.

Respiratory sensitisation

Based on available data, the classification criteria are not met.

Product:

Test Type	:	Maximisation Test
Species	:	Guinea pig
Result	:	Animal test did not cause sensitization by skin contact.
GLP	:	yes

Components:

azimsulfuron (ISO):

Test Type	:	Maximisation Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Animal test did not cause sensitization by skin contact.
GLP	:	yes
Remarks	:	Information source: Internal study report

Talc (Mg3H2(SiO3)4):

Test Type	:	Maximisation Test
Exposure routes	:	Dermal
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Does not cause skin sensitisation.

Exposure routes	:	Inhalation
Species	:	Rat
Result	:	Does not cause respiratory sensitisation.

sodium benzoate:

Test Type	:	Local lymph node assay (LLNA)
Species	:	Mouse
Result	:	Does not cause skin sensitisation.

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Product:

Genotoxicity in vitro	:	Test Type: Ames test Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative
Genotoxicity in vivo	:	Test Type: Micronucleus test Result: negative

Components:

azimsulfuron (ISO):

Genotoxicity in vitro	:	Test Type: reverse mutation assay Test system: Salmonella typhimurium Metabolic activation: with and without metabolic activation Result: negative Test Type: reverse mutation assay Test system: Escherichia coli Metabolic activation: with and without metabolic activation Result: negative Test Type: unscheduled DNA synthesis assay Test system: rat hepatocytes Method: OECD Test Guideline 482 Result: negative
Genotoxicity in vivo	:	Test Type: Micronucleus test Species: mice (male and female) Application Route: Ingestion Method: OECD Test Guideline 474 Result: negative
Germ cell mutagenicity - Assessment	:	Animal testing did not show any mutagenic effects.

Talc (Mg3H2(SiO3)4):

Genotoxicity in vitro	:	Test Type: In vitro mammalian cell gene mutation test Result: negative Test Type: gene mutation test Method: QSAR Result: negative Test Type: reverse mutation assay Result: negative
Genotoxicity in vivo	:	Test Type: dominant lethal test Species: Rat (male) Application Route: Oral Result: negative

Germ cell mutagenicity - Assessment	:	Weight of evidence does not support classification as a germ cell mutagen.

magnesium carbonate:

Genotoxicity in vitro	:	Test Type: reverse mutation assay Method: OECD Test Guideline 471 Result: negative Remarks: Based on data from similar materials Test Type: Chromosome aberration test in vitro Method: OECD Test Guideline 473 Result: negative Remarks: Based on data from similar materials Test Type: gene mutation test Result: negative Remarks: Based on data from similar materials

sodium benzoate:

Genotoxicity in vitro	:	Test Type: reverse mutation assay Method: OECD Test Guideline 471 Result: negative
Genotoxicity in vivo	:	Test Type: chromosome aberration assay Species: Rat (male) Application Route: Ingestion Method: OECD Test Guideline 475 Result: negative

Carcinogenicity

Based on available data, the classification criteria are not met.

Components:

azimsulfuron (ISO):

Species	:	Rat, male and female
Application Route	:	Ingestion
Exposure time	:	24 month(s)
Method	:	OECD Test Guideline 453
Result	:	negative
Carcinogenicity - Assessment	:	Did not show carcinogenic effects in animal experiments.

Talc (Mg3H2(SiO3)4):

Species	:	Rat, male and female
Application Route	:	Oral
Exposure time	:	101 days
Dose	:	100 mg/kg bw/day
NOAEL	:	100 mg/kg bw/day
Method	:	OECD Test Guideline 453

Result	:	negative
Target Organs	:	Stomach
Tumor Type	:	Leiomyosarcoma

Carcinogenicity - Assessment : Weight of evidence does not support classification as a carcinogen

magnesium carbonate:

Species	:	Mouse, male and female
Application Route	:	Oral
Exposure time	:	96 weeks
Dose	:	0, 0.5 and 2% MgCl2.6H2O
Method	:	OECD Test Guideline 451
Remarks	:	Based on data from similar materials

Carcinogenicity - Assessment : Animal testing did not show any carcinogenic effects.

sodium benzoate:

Species	:	Rat, male and female
Application Route	:	Oral
Exposure time	:	730 d
Result	:	negative

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

azimsulfuron (ISO):

Effects on fertility : Test Type: Two-generation study
Species: Rat, male and female
Application Route: Ingestion
General Toxicity - Parent: NOEL: 125 ppm
Fertility: NOEL: 8.000 ppm
Method: OECD Test Guideline 416
Result: negative

Effects on foetal development : Test Type: Embryo-foetal development
Species: Rat
Application Route: Ingestion
General Toxicity Maternal: NOEL: 200 mg/kg bw/day
Teratogenicity: NOEL: 1.000 mg/kg bw/day
Symptoms: Maternal effects
Method: EPA OPP 83-3
Result: negative

Reproductive toxicity - Assessment : Animal testing did not show any effects on fertility.
Did not show teratogenic effects in animal experiments.

Talc (Mg3H2(SiO3)4):

Effects on fertility : Species: Rabbit, female

		Application Route: Oral Dose: 9, 42, 195, 900 mg/kg bw/day General Toxicity - Parent: NOAEL: > 900 mg/kg body weight General Toxicity F1: NOAEL: > 900 mg/kg body weight Result: negative
Effects on foetal development	:	Test Type: reproductive and developmental toxicity study Species: Rat Application Route: Oral Dose: 0,16,74,350,1600mg/kg bw/day Duration of Single Treatment: 20 d General Toxicity Maternal: NOAEL: >= 1.600 mg/kg bw/day Embryo-foetal toxicity: NOAEL: 1.600 mg/kg bw/day Result: negative
Reproductive toxicity - Assessment	:	Weight of evidence does not support classification for reproductive toxicity

magnesium carbonate:

Effects on fertility	:	Test Type: reproductive and developmental toxicity study Species: Rat, male and female Application Route: Oral Dose: 0, 250, 500 and 1000 mg/kg bw Duration of Single Treatment: 28 - 31 d Method: OECD Test Guideline 422 Result: negative Remarks: Based on data from similar materials
Effects on foetal development	:	Test Type: reproductive and developmental toxicity study Species: Rat, male and female Application Route: Oral Dose: 0, 250, 500 and 1000 mg/kg bw Duration of Single Treatment: 28 - 54 d Developmental Toxicity: NOAEL: 1.000 mg/kg body weight Method: OECD Test Guideline 422 Remarks: Based on data from similar materials Test Type: reproductive and developmental toxicity study Species: Rat Application Route: Oral Dose: 0, 200, 400 and 800 mg/kg/day Duration of Single Treatment: 14 d Teratogenicity: NOAEL: > 800 mg/kg body weight Method: OECD Test Guideline 414 Remarks: Based on data from similar materials
Reproductive toxicity - Assessment	:	Weight of evidence does not support classification for reproductive toxicity

sodium benzoate:

Effects on fertility	:	Test Type: Two-generation study Species: Rat, male and female Application Route: Oral Result: negative

Effects on foetal development	:	Test Type: reproductive and developmental toxicity study Species: Rat Application Route: Ingestion Method: OECD Test Guideline 414 Result: negative

STOT - single exposure

Based on available data, the classification criteria are not met.

Components:

azimsulfuron (ISO):

Assessment	:	The substance or mixture is not classified as specific target organ toxicant, single exposure.

Talc ($Mg_3H_2(SiO_3)_4$):

Assessment	:	The substance or mixture is not classified as specific target organ toxicant, single exposure.

STOT - repeated exposure

Based on available data, the classification criteria are not met.

Components:

azimsulfuron (ISO):

Assessment	:	The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

magnesium carbonate:

Assessment	:	The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

Components:

azimsulfuron (ISO):

Species	:	Rat, male
NOAEL	:	75,3 mg/kg
Application Route	:	Oral
Exposure time	:	90 d
Method	:	OECD Test Guideline 408
GLP	:	yes

Species	:	Rat, female
NOAEL	:	82,4 mg/kg
Application Route	:	Oral
Exposure time	:	90 d
Method	:	OECD Test Guideline 408
GLP	:	yes

Talc (Mg3H2(SiO3)4):

Species	:	Rat, male and female
NOAEL	:	100 mg/kg
Application Route	:	Oral - feed
Exposure time	:	101 d
Dose	:	100 mg/kg bw/day

Species	:	Rat, male and female
NOAEL	:	2 mg/m3
LOAEL	:	6 mg/m3
Application Route	:	inhalation (dust/mist/fume)
Test atmosphere	:	dust/mist
Exposure time	:	20 d
Dose	:	0, 2, 6, 18 mg/m³

magnesium carbonate:

Species	:	Rat, male and female
NOAEL	:	1.000 mg/kg
Application Route	:	Oral
Exposure time	:	28 - 31d
Dose	:	0, 250, 500 and 1000 mg/kg bw/
Method	:	OECD Test Guideline 422
Remarks	:	Based on data from similar materials

Species	:	Rat, male
NOAEL	:	308 mg/kg
Application Route	:	Oral
Exposure time	:	90d
Dose	:	0, 62, 308,1600 mg MgCl2/kg bw
Method	:	OECD Test Guideline 408
Remarks	:	Based on data from similar materials

Species	:	Rat, female
LOAEL	:	299 mg/kg
Application Route	:	Oral
Exposure time	:	90d
Dose	:	0, 59, 299,1531 mg MgCl2/kg
Method	:	OECD Test Guideline 408
Remarks	:	Based on data from similar materials

Species	:	Mouse, male and female
Application Route	:	Oral
Exposure time	:	13 weeks
Dose	:	2.5%
Method	:	OECD Test Guideline 408
Target Organs	:	Kidney
Remarks	:	Based on data from similar materials

sodium benzoate:

Species	:	Rat, male and female
NOAEL	:	1.000 mg/kg
Application Route	:	Oral - feed

Aspiration toxicity

Based on available data, the classification criteria are not met.

Components:

azimsulfuron (ISO):

The substance does not have properties associated with aspiration hazard potential.

Further information

Product:

Remarks : No data available

12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

Toxicity to fish	:	LC50 (Oncorhynchus mykiss (rainbow trout)): 492 mg/l Exposure time: 96 h Test Type: static test Method: OECD Test Guideline 203
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): > 1.000 mg/l Exposure time: 48 h Test Type: static test Method: OECD Test Guideline 202
Toxicity to algae/aquatic plants	:	ErC50 (Pseudokirchneriella subcapitata (green algae)): 0,075 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 EbC50 (Pseudokirchneriella subcapitata (green algae)): 0,015 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 Remarks: (Data on the product itself) Information source: Internal study report
Toxicity to soil dwelling organisms	:	LC50 (Eisenia fetida (earthworms)): > 1.000 mg/kg Exposure time: 14 d Remarks: No significant adverse effect on nitrogen mineralization. No significant adverse effect on carbon mineralization.
Toxicity to terrestrial organisms	:	LD50 (Apis mellifera (bees)): > 349,6 µg/bee End point: Acute oral toxicity LD50 (Apis mellifera (bees)): > 400 µg/bee End point: Acute contact toxicity

LD50 (Colinus virginianus (Bobwhite quail)): > 5620 ppm
Method: OECD Test Guideline 205

LD50 (Anas platyrhynchos (Mallard duck)): > 5620 ppm
Method: OECD Test Guideline 205

Components:

azimsulfuron (ISO):

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 154 mg/l
Exposure time: 96 h
Test Type: static test
Method: OECD Test Guideline 203
GLP: yes
Remarks: Information source: Internal study report

LC50 (Lepomis macrochirus (Bluegill sunfish)): > 1.000 mg/l
Exposure time: 96 h
Test Type: static test
Method: OECD Test Guideline 203
GLP: yes
Remarks: Information source: Internal study report

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 600 mg/l
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202
GLP: yes
Remarks: Information source: Internal study report

Toxicity to algae/aquatic plants : EbC50 (Pseudokirchneriella subcapitata (green algae)): 0,012 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
GLP: yes

ErC50 (Pseudokirchneriella subcapitata (green algae)): 0,099 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
GLP: yes

EC50 (Lemna gibba (duckweed)): 0,00062 mg/l
End point: Frond
Exposure time: 7 d
Test Type: Static renewal test
Method: OECD Test Guideline 221

NOEC (Lemna gibba (duckweed)): 0,00019 mg/l
End point: Frond
Exposure time: 7 d
Test Type: Static renewal test
Method: OECD Test Guideline 221

M-Factor (Acute aquatic tox- : 10

icity)		
Toxicity to fish (Chronic toxicity)	:	NOEC (Oncorhynchus mykiss (rainbow trout)): 23 mg/l Exposure time: 28 d Test Type: flow-through test Method: OECD Test Guideline 204 GLP: yes Remarks: Information source: Internal study report NOEC (Oncorhynchus mykiss (rainbow trout)): 6,3 mg/l Exposure time: 90 d Method: OECD Test Guideline 210 GLP: yes Remarks: Information source: Internal study report
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC (Daphnia magna (Water flea)): 5,4 mg/l Exposure time: 21 d Method: OECD Test Guideline 202 GLP: yes Remarks: Information source: Internal study report
M-Factor (Chronic aquatic toxicity)	:	100
Toxicity to soil dwelling organisms	:	LC50 (Eisenia fetida (earthworms)): > 1.000 mg/kg Exposure time: 14 d Method: OECD Test Guideline 207 GLP: yes Remarks: Information source: Internal study report NOEC (Eisenia fetida (earthworms)): 12,5 mg/kg End point: reproduction Method: OECD Test Guideline 222 GLP: yes
Toxicity to terrestrial organisms	:	LD50 (Colinus virginianus (Bobwhite quail)): > 2.250 mg/kg Method: US EPA Test Guideline OPP 71-1 GLP: yes Remarks: Information source: Internal study report LC50 (Anas platyrhynchos (Mallard duck)): > 5.620 mg/kg Exposure time: 8 d Method: OECD Test Guideline 205 GLP: yes Remarks: Information source: Internal study report LD50 (Apis mellifera (bees)): > 0,025 mg/kg Method: US EPA Test Guideline OPP 141-1 GLP: yes Remarks: Contact, Information source: Internal study report LD50 (Apis mellifera (bees)): > 1.000 mg/kg Method: US EPA Test Guideline OPP 141-1 GLP: yes Remarks: Dietary, Information source: Internal study report

		NOEC (Anas platyrhynchos (Mallard duck)): 172 mg/kg End point: Reproduction Test Method: OECD Test Guideline 206 GLP: yes

Talc (Mg3H2(SiO3)4):

Toxicity to fish	:	LC50 (Fish): 89.581,016 mg/l Exposure time: 96 h Method: QSAR
Toxicity to daphnia and other aquatic invertebrates	:	LC50 (Daphnia magna (Water flea)): 36.812,359 mg/l Exposure time: 48 h Method: QSAR
Toxicity to algae/aquatic plants	:	NOEC (green algae): 918,089 mg/l Exposure time: 30 d Method: QSAR EC50 (green algae): 7.202,7 mg/l Exposure time: 96 h Method: QSAR
Toxicity to fish (Chronic toxicity)	:	NOEC (Fish): 1.412,648 mg/l Exposure time: 30 d Method: QSAR
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC (Daphnia (water flea)): 1.459,798 mg/l Exposure time: 30 d Method: QSAR

Lignosulfonic acid, Sodium salt:

Toxicity to fish	:	EC50 (Danio rerio (zebra fish)): > 1.000 mg/l Exposure time: 96 h Remarks: Based on data from similar materials
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): > 1.000 mg/l Exposure time: 48 h Remarks: Based on data from similar materials
Toxicity to algae/aquatic plants	:	EC50 (Scenedesmus subspicatus): > 600 mg/l Exposure time: 72 h Remarks: Based on data from similar materials

magnesium carbonate:

Toxicity to fish	:	LC50 (Pimephales promelas (fathead minnow)): 2.120 mg/l Exposure time: 96 h Test Type: static test Remarks: Based on data from similar materials
Toxicity to daphnia and other aquatic invertebrates	:	LC50 (Daphnia magna (Water flea)): 140 mg/l Exposure time: 48 h Test Type: static test

		Remarks: Based on data from similar materials
Toxicity to algae/aquatic plants	:	EC50 (Desmodesmus subspicatus (green algae)): > 18,5 mg/l Exposure time: 72 h Test Type: static test Method: OECD Test Guideline 201 Remarks: Based on data from similar materials NOEC (Desmodesmus subspicatus (green algae)): 18,5 mg/l Exposure time: 72 h Test Type: static test Method: OECD Test Guideline 201 Remarks: Based on data from similar materials
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	Exposure time: 21 d Test Type: semi-static test Remarks: Based on data from similar materials
Toxicity to microorganisms	:	EC50 (activated sludge): > 900 mg/l Exposure time: 3 h Test Type: Respiration inhibition Method: OECD Test Guideline 209 Remarks: Based on data from similar materials

sodium benzoate:

Toxicity to fish	:	LC50 (Pimephales promelas (fathead minnow)): 484 mg/l Exposure time: 96 h Method: EPA OPP 72-1
Toxicity to daphnia and other aquatic invertebrates	:	LC50 (Daphnia magna (Water flea)): > 100 mg/l Exposure time: 96 h
Toxicity to algae/aquatic plants	:	EC50 (Pseudokirchneriella subcapitata (green algae)): > 30,5 mg/l Exposure time: 72 h Method: OECD Test Guideline 201
Toxicity to fish (Chronic toxicity)	:	NOEC (Danio rerio (zebra fish)): 10 mg/l Exposure time: 6 d
Toxicity to microorganisms	:	NOEC (sewage treatment plant microorganisms): > 100 mg/l Exposure time: 168 h

Persistence and degradability

Components:

azimsulfuron (ISO):

Biodegradability	:	Biodegradation: 12 % Exposure time: 28 d Method: OECD Test Guideline 301E GLP: yes Remarks: Not readily biodegradable.

Stability in water	:	Degradation half life: 89 d (25 °C) pH: 5 Degradation half life: 132 d (25 °C) pH: 9
Photodegradation	:	

Lignosulfonic acid, Sodium salt:

Biodegradability	:	Result: Not readily biodegradable. Remarks: Based on data from similar materials

sodium benzoate:

Biodegradability	:	Result: Readily biodegradable. Biodegradation: >= 50 % Exposure time: 60 d Method: OECD Test Guideline 311

Bioaccumulative potential

Components:

azimsulfuron (ISO):

Bioaccumulation	:	Remarks: Bioaccumulation is unlikely.
Partition coefficient: n-octanol/water	:	log Pow: -1,36

Talc (Mg3H2(SiO3)4):

Bioaccumulation	:	Bioconcentration factor (BCF): 3,16 Method: QSAR
Partition coefficient: n-octanol/water	:	log Pow: -9,4 (25 °C) pH: 7 Method: QSAR

sodium benzoate:

Bioaccumulation	:	Remarks: Bioaccumulation is unlikely.
Partition coefficient: n-octanol/water	:	log Pow: 1,88 Remarks: Based on data from similar materials

Mobility in soil

Components:

azimsulfuron (ISO):

Distribution among environmental compartments	:	Koc: 85,34 - 142,56 ml/g Remarks: Moderately mobile in soil

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

Hygienic standards:

(Allowable concentration in air, water, including fishery waters, soil)

Components	Air	Water	Soil	Data Source
Talc (Mg3H2(SiO3)4) 14807-96-6	TSEL: 0,5 mg/m3	TSEL: 0,25 mg/l Limiting health hazard indicator: organoleptic; increases the turbidity of the water Hazard class: Class 4 - low hazard	No data available	List 2 List 3
Lignosulfonic acid, Sodium salt 8061-51-6	TSEL: 0,5 mg/m3	MPC: 3 Milligrams per cubed decimeter Limiting health hazard indicator: sanitary and toxicological effects Hazard class: 4 MPC: 3 Milligrams per cubed decimeter Limiting health hazard indicator: toxic Hazard class: 4	No data available	List 2 List 5
sodium benzoate 532-32-1	TSEL: 0,05 mg/m3	TSEL: 0,1 mg/l Limiting health hazard indicator: general sanitary Hazard class: Class 3 - moderately dangerous	No data available	List 2 List 3

For explanation of abbreviations see section 16.

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemi-

		cal or used container. Send to a licensed waste management company.
Contaminated packaging	:	Empty remaining contents. Triple rinse containers. Do not re-use empty containers. Packaging that is not properly emptied must be disposed of as the unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. TRANSPORT INFORMATION

ADR

UN number	:	UN 3077
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Azimsulfuron)
Class	:	9
Packing group	:	III
Labels	:	9
Hazard Identification Number	:	90
Tunnel restriction code	:	(-)
Environmentally hazardous	:	yes

UNRTDG

UN number	:	UN 3077
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Azimsulfuron)
Class	:	9
Subsidiary risk	:	ENVIRONM.
Packing group	:	III
Labels	:	9 (ENVIRONM.)
Environmentally hazardous	:	yes

IATA-DGR

UN/ID No.	:	UN 3077
Proper shipping name	:	Environmentally hazardous substance, solid, n.o.s. (Azimsulfuron)
Class	:	9
Packing group	:	III
Labels	:	Miscellaneous
Packing instruction (cargo aircraft)	:	956
Packing instruction (passenger aircraft)	:	956
Environmentally hazardous	:	yes

IMDG-Code

UN number	:	UN 3077
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Azimsulfuron)
Class	:	9

Packing group	:	III
Labels	:	9
EmS Code	:	F-A, S-F
Marine pollutant	:	yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

The components of this product are reported in the following inventories:

TCSI	:	Not in compliance with the inventory
TSCA	:	Product contains substance(s) not listed on TSCA inventory.
AIIC	:	Not in compliance with the inventory
DSL	:	This product contains chemical substance(s) exempt from CEPA DSL Inventory requirements. It is regulated as a pesticide subject to Pest Control Products Act (PCPA) requirements. Read the PCPA label, authorized under the Pest Control Products Act, prior to using or handling this pest control product.
ENCS	:	Not in compliance with the inventory
ISHL	:	Not in compliance with the inventory
KECI	:	On the inventory, or in compliance with the inventory
PICCS	:	Not in compliance with the inventory
IECSC	:	Not in compliance with the inventory
NZIoC	:	Not in compliance with the inventory
TECI	:	Not in compliance with the inventory

16. OTHER INFORMATION

Full text of H-Statements

H303	May be harmful if swallowed.

H313	May be harmful in contact with skin.
H319	Causes serious eye irritation.
H400	Very toxic to aquatic life.
H402	Harmful to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	:	Acute toxicity
Aquatic Acute	:	Short-term (acute) aquatic hazard
Aquatic Chronic	:	Long-term (chronic) aquatic hazard
Eye Irrit.	:	Eye irritation
List 2	:	SanPiN 1.2.3685-21 Table 1.2, Table 1.12 & Table 1.13 Tentative Safe Exposure Levels (TSEL) in the air of urban and rural settlements
List 3	:	SanPiN 1.2.3685-21 Table 3.14 & Table 3.18 Indicative permissible levels (TAC) of chemicals in the water of drinking systems of centralized, including hot, and non-centralized water supply, water of ground and surface water bodies of drinking and cultural and domestic water use, water of swimming pools, water parks
List 5	:	Order of the Russian Federal Fisheries Agency "Standards of maximum permissible concentrations of harmful substances in fishery water bodies"

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

FMC Corporation believes that the information and recommendations contained herein (including data and statements) are accurate as of the date hereof. You can contact FMC Corporation to ensure that this document is the most current available from FMC Corporation. No warranty of fitness for any particular purpose, warranty of merchantability or any other warranty, expressed or implied, is made concerning the information provided herein. The information provided herein relates only to the specified product designated and may not be applicable where such product is used in combination with any other materials or in any process. The user is responsible for determining whether the product is fit for a particular purpose and suitable for the user's conditions and methods of use. Since the conditions and methods of use are beyond the control of FMC Corporation, FMC Corporation expressly disclaims any and all liability as to any results obtained or arising from any use of the products or reliance on such information.

TM / 6N